

Seungmin Yome

M.S. Student in Energy & Petroleum Engineering

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Research Experience

University of Wyoming, Graduate Research Assistant

Laramie, WY, USA
2025 – present

- Built the data generator: an axisymmetric finite-volume model (OpenFOAM + solids4Foam) of a 1 mm ball pressed into and unloaded from Mohr–Coulomb rock; measured rock inputs are published Montney strength values, none adjusted to indentation data
- Checked it on three finer meshes against published Brinell tests (22 samples, 35 N): peak depth 15.2–17.1 μm vs. $15.2 \pm 2.4 \mu\text{m}$ measured, not yet converged; a Hertz elastic benchmark recovered only 79% of the load
- Predicted eight published proppant-pack tests blind (69% mean absolute error) and traced the misfit to grain size: $R^{1.0}$ scaling in the model vs. $R^{0.47}$ measured, unchanged by mesh or dilation
- Found that a one-group correlation ($h/R = 0.0031 \Pi^{0.72}$, $R^2 = 0.88$ on 8 runs) collapses to $R^2 = 0.11$ once rock properties vary, so a single dimensionless group cannot carry the law
- Replaced it with a joint surrogate: peak depth as a power law in normalized load, modulus ratio, compliance ratio, $\tan \phi$ and $\tan \psi$; end depth as peak times a logistic recovery ratio; 1.45% / 2.29% nested-CV error, stated as a surrogate, not a validated law
- Showed by sweeps that the input this law needs most is the one nobody has measured: a 10° dilation angle cuts the end depth 45% and moves the retained fraction from 67% to 44% (measured $\approx 40\%$)
- So the dilation angle became an inverse-ML problem: kNN, gradient-boosted trees, TabPFN and a 1-D CNN on ≈ 250 new simulations with split-conformal 90% intervals, to learn which indentation measurement pins it down
- Designed and built the rig for those measurements: displacement-controlled micro-indentation, 3D-printed fixtures through up to 20 revisions, 445 N load cell (0.004 N) and laser vibrometer at 2 kHz via NI DAQ/LabVIEW; fixed their cross-talk
- Run every campaign through an AI coding agent (Claude Code) with test plans and pass/fail rules committed to Git before any run: 426 runs, $\approx 1,400$ core-hours, every manuscript number traceable to a hashed table and script
- Put the manuscript through a dozen adversarial AI review passes: retracted a headline -0.8% validation figure, disclosed the failed elastic benchmark, and separated the computed end depth from the physical residual depth

KAIST — Graduate School of Mechanical Engineering, Research Intern (6-month full-time)

South Korea
2024 – 2025

- Advisor: Dr. Yeunwoo Cho
- Sole investigator on a government-funded study of air-jet-driven waves on a thin elastic film: derived the dispersion relation with film stiffness and tension and confirmed the Kelvin-wave transition on high-speed video
- Assembled a free-surface cavitation tank from parts and explained the upper-wall contraction of ventilated supercavities with a wake-theory term in a 2D model's closure condition, closing most of the gap to the photos
- Simulated von Kármán vortex shedding near a free surface (ANSYS Fluent, VOF): the closer the surface, the lower the Strouhal number and the weaker the upper vortices

Pusan National University — Department of Naval Architecture & Ocean Engineering,

Undergraduate Research Assistant

South Korea

2022 – 2024

- First CFD project: validated KCS hull resistance against the Gothenburg 2010 benchmark (STAR-CCM+, ≈ 4.4 M cells) to 1.16%; owned the mesh, refining bow, stern and free surface after every run
- PIV and hull-resistance tests, ITTC 1957/1978 propulsion design, and an ANSYS Fluent study of an airfoil appendage on hull attitude and resistance
- Started the vibration/acoustic anomaly-detection model (MATLAB, Python) finished at Samsung

Industry Experience

Samsung Heavy Industries Laboratory, Mechanical Engineer Intern

South Korea

2024 – 2024

- Trained an LSTM autoencoder on healthy-condition vibration and acoustic data only (Green Nuri vessel program): 621 of 622 anomalies caught, 8 false alarms, 99.49% accuracy
- One flag traced to a fatigue crack along a weld; the model saw that something changed but not why, which is what pulled me toward smaller-scale mechanics

Pusan National University, Capstone Design Project

South Korea

2023 – 2023

- Benchmarked engineering design tools for the shipbuilding industry (AVEVA Marine, TTM, VATIA, AutoCAD) across performance, integration, and usability criteria
- Proposed a cloud-based ML-integrated design platform (Google Cloud) balancing system scalability with market viability

Publications

Joint prediction of peak proppant embedment and post-unloading residual depth: a dimensionless candidate model and its verification status

in preparation

Seungmin Yome, Soheil Saraji

(Acta Geotechnica)

Numerical Recovery of Hydroelastic Wave Dispersion in Coupled Elastic Film-Liquid Systems Using a Finite-Volume Fluid-Structure Interaction Solver

in preparation

Seungmin Yome

(Journal of Fluids and Structures)

Skills

Simulation & CAE

Programming

Design & Engineering Tools

Laboratory

Education

University of Wyoming, Energy & Petroleum Engineering

Laramie, WY, USA

2025 – present

- Advisor: Dr. Soheil Saraji (Department Head)
- Minor: Computer Science (in progress)
- Thesis: Numerical modeling of proppant-supported fracture systems — embedment, rock deformation, and single-phase fluid transport using solids4Foam
- Coursework: Rock Mechanics, Enhanced Oil Recovery, Geostatistics, Python for Engineers
- Current coursework (Fall 2026): ML for Computational Solid Mechanics, Practical Machine Learning

Pusan National University , Naval Architecture & Ocean Engineering • Advisor: Dr. Inwon Lee • Coursework: Solid Mechanics, Fluid Mechanics, Linear Algebra, Thermodynamics, Ocean Wave Mechanics, FEM, Offshore Structural Design, CFD	South Korea 2018 – 2024
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Grants & Fellowships

Graduate Research Assistant, University of Wyoming (Full-funded Graduate Research Assistantship) • University of Wyoming	2025–Present
Research Scholarship • Korea Advanced Institute of Science and Technology	Jan 2024
Undergraduate Research Student Grants • Pusan National University	2022–2024
Academic Excellence Scholarship • Pusan National University	2022–2024
Foreign Language Special Lecture Consignment Program Scholarship • Pusan National University	Jan 2021
Creative Human Resources Development Project Group Scholarship • Pusan National University	Jan 2018

Selected Activities & Awards

Invited to Samsung Heavy Industries Wind Propulsion Ship Forum (travel award) • Samsung Heavy Industries	July 2024
Participating in ‘Conceptual Design of Efficient Devices of a 25-tonne High-Speed Ship’ • Korean Society of Shipbuilding and Marine Engineering / Coast Guard	Jan. 2024
Invited to Hyundai Heavy Industries Eco-Friendly Smart Ship Forum (travel award) • Hyundai Heavy Industries	Apr. 2023
TPC (Test Preparation Course) Certification — English Consignment Program • Pusan National University	July 2021

Teaching & Tutoring

Private Tutoring (Educational Volunteering) , Mathematics Instructor • Taught high school mathematics to undergraduate-level students as part of educational volunteering activities over two years	South Korea 2022 – 2024
Private Tutoring , Physics & Chemistry Instructor — College Entrance Preparation • Taught physics and chemistry to 10 high school seniors (Grade 12) preparing for the Korean college entrance exam (CSAT) over a 6-month intensive period • 9 out of 10 students (90%+) gained admission to top-tier Korean universities	South Korea 2023 – 2024

Languages

Korean
Native

English
Fluent